

Airline Booking State

Define the airline state schema, which holds the set of routes and the bookings relation.

$[Airline, BookingId, CustomerId, RouteId]$

<i>AirlineState</i>
$routes : \mathbb{P} RouteId$
$bookings : BookingId \leftrightarrow CustomerId \times RouteId$

A read-only operation that queries whether a route exists uses Xi to signal that the state is unchanged.

<i>RouteExists</i>
$\exists AirlineState$
$routeId? : RouteId$
$result! : \mathbb{N}$
$routeId? \in routes$
$result! = 1$

An operation that adds a booking uses Delta to include both the before- state and the after-state. The primes on bookings' and routes' are Z convention for the output state; here bookings' is added as a typed declaration because the assignment is the predicate.

<i>AddBooking</i>
$\Delta AirlineState$
$bookingId? : BookingId$
$customerId? : CustomerId$
$routeId? : RouteId$
$bookingId? \notin \text{dom } bookings$
$routeId? \in routes$

Stack with Generic Inclusion

Schema inclusion works with generic types. Delta Stack[N] brings in the before-and-after state of a stack instantiated over the naturals.

<i>StackOp</i> [X]
$delta : \mathbb{N}$

<i>PushNat</i>
$\Delta StackOp[\mathbb{N}]$
$value? : \mathbb{N}$
$delta > 0$